



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Internet of Things

IoT-Based Environmental Monitoring System with Digital Twin and Real Time Web Dashboard Project Report

Submitted to:

Dr. Amanpreet Kaur
Assistant Professor, Dept. of ECE

Submitted by:

Abhinivesh Sharma (102306168)(3F23)

Devansh Gaira (102306239)(3F23)

Course: Internet of Things (UEC640)

Department of Electronics and Communication Engineering
Thapar Institute of Engineering and Technology

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Abstract

Affordable IoT sensor systems used for environmental monitoring often suffer from inaccuracies due to electrical noise, hysteresis effects, and long-term drift. These limitations reduce the reliability of low-cost deployments, especially in real-world conditions where environmental influences introduce additional bias.

This project addresses these challenges by developing a complete IoT system augmented with a software-based digital twin. The system collects local environmental data and integrates it with external atmospheric references. The digital twin processes incoming sensor data using adaptive correction, temporal modeling, and predictive estimation.

Unlike traditional multi-stage pipelines, the implemented twin uses a feedback-driven adaptive model that continuously refines sensor output. The processed data is visualized through a real-time Streamlit dashboard, enabling monitoring, anomaly detection, and environmental interpretation.

This approach demonstrates that significant improvements in data quality can be achieved through software without modifying hardware.

1 Introduction

The Internet of Things (IoT) has significantly expanded environmental monitoring capabilities through the deployment of distributed sensing systems. With the availability of low-cost microcontrollers such as the ESP32 and MEMS-based sensors, it is now possible to deploy monitoring nodes at high spatial density.

However, the reliability of such systems remains a major challenge. Sensor readings are affected by electrical noise, ADC quantization, thermal drift, and environmental influences. These factors introduce both random fluctuations and systematic bias.

In practical scenarios, sensors are also influenced by local effects such as solar heating and airflow. Gas sensors exhibit additional complexities such as humidity sensitivity and baseline drift over time. As a result, raw sensor readings often deviate significantly from actual environmental conditions.

Most undergraduate implementations ignore these issues and simply display raw sensor outputs. This limits their usefulness in real-world applications.

This project addresses these limitations by introducing a digital twin architecture that enhances sensor data using adaptive modeling and real-time feedback.

2 Problem Statement

Low-cost IoT sensor nodes often produce unreliable data due to electrical noise, hysteresis, environmental effects, and long-term drift. Existing implementations typically apply isolated corrections and lack a unified framework for handling these errors. This project addresses this limitation by developing an integrated system combining hardware sensing with a digital twin for real-time adaptive correction and reliable data interpretation.

3 Literature Review

IoT-based environmental monitoring has grown significantly over the past decade, driven by the availability of low-cost, WiFi-capable microcontrollers such as the ESP32. Several research groups have built sensor nodes using this platform.

3.1 ESP32-Based IoT Monitoring Systems

One representative direction in recent literature is the use of ESP32 as a standalone environmental data node with cloud connectivity.

An IEEE conference paper on factory environmental monitoring using ESP32-S3 demonstrated autonomous data collection, analysis, and visualization through graph signal processing, achieving reliable operation with low cost and convenient deployment [1].

Another study specifically targeting air quality monitoring with ESP32 measured PM2.5, PM10, CO, NO₂, SO₂, and O₃, reporting an R^2 value of 0.9402 for PM2.5 regression against reference equipment [2].

A third IEEE contribution evaluated the energy consumption of the ESP32 under four different operating conditions using the MQTT protocol, which is directly relevant to understanding the power budget of continuous-monitoring deployments [3].

What these systems share is a straightforward pipeline: sensors are read, data is transmitted to a cloud service or displayed locally, and the raw values are presented to the user without any compensation for the known error sources in the sensors themselves.

None of them develop a software model that tracks sensor state or applies physics-informed corrections.

3.2 BMP280 Sensor Characteristics and Limitations

The Bosch Sensortec BMP280 is among the most widely used MEMS sensors for combined temperature and pressure measurement in IoT systems.

Its datasheet specifies temperature accuracy of $\pm 1^{\circ}\text{C}$ and absolute pressure accuracy of ± 1 hPa under normal operating conditions [4].

A well-documented limitation observed in practice is that when the BMP280 is mounted close to a microcontroller on a shared breadboard or PCB, the heat dissipated by the processor introduces a positive temperature offset in the sensor reading, typically in the range of 1°C to 2°C .

This has been reported by multiple practitioners working with the Adafruit BMP280 breakout board and the ESP32 in particular, and it represents a systematic bias that simple averaging cannot remove.

The bilinear feedback stage in this project's digital twin is designed to partially address this effect by weighting the external reference temperature from the OpenWeatherMap API alongside the local reading.

3.3 MQ135 Gas Sensor: Sensitivity, Humidity, and Calibration

The MQ135 is a metal oxide semiconductor (MOS) gas sensor manufactured by Zhengzhou Winsen Electronics.

Its datasheet specifies sensitivity to ammonia, benzene, sulphide, NO_x , CO_2 , smoke, and other toxic gases, with a characteristic sensitivity curve measured at 20°C and 65% relative humidity [5].

A key published limitation is that the output resistance of MQ-series sensors drifts with changes in temperature and humidity.

A study specifically examining the influence of temperature and humidity on the output resistance ratio of the MQ-135 found that these environmental factors introduce measurable errors in gas concentration readings, concluding that compensation techniques are necessary for reliable quantitative results [6].

A separate study on temperature and humidity compensation for MOS gas sensors using Random Forest regression demonstrated approximately 1% improvement in accuracy over uncompensated readings when using algorithmic correction [7].

In this project, a simpler linear humidity correction is applied, dividing the raw ADC value by

a factor of:

$$AQ_{corr} = \frac{AQ_{raw}}{1 + 0.005(RH - 50)}$$

which approximates the known sensitivity drift described in the MQ135 datasheet.

An important limitation that none of the reviewed studies fully resolve is the absence of cross-sensitivity separation: the MQ135 responds to multiple gases simultaneously and cannot isolate individual gas concentrations without additional sensors or chemometric methods.

3.4 Digital Twin Concepts and IoT Applications

The concept of a digital twin was formally introduced by Grieves and Vickers (2017) as a virtual information construct that fully describes a physical system, with the explicit goal of mitigating unpredictable emergent behaviour by maintaining a synchronized software model [9].

Tao et al. (2018) extended this framework to industrial informatics, defining a digital twin as comprising the physical entity, the virtual model, and the data connection between them, and identifying sensor health monitoring and prognostics as among the most valuable applications [10].

Fuller et al. (2020) reviewed the enabling technologies, challenges, and open research questions for digital twins in IoT contexts, noting that lightweight edge-deployable twins that operate without cloud infrastructure remain an underexplored area [11].

These foundational works establish the conceptual basis for the software layer in this project.

The key distinction between a digital twin and a simple data logger is that the twin maintains a model with internal state, applies physics-informed transformations to raw sensor data, and produces outputs such as predictions and health scores that the raw sensor cannot provide.

4 Hardware Components

The hardware layer of the system is centered around the ESP32 development board, a dual-core microcontroller operating at up to 240 MHz with integrated WiFi and Bluetooth capabilities. The ESP32 is selected for its high computational capability, low power consumption, and ability to interface with multiple sensors simultaneously.

The primary sensing component for environmental data acquisition is the BMP280 sensor,

which provides temperature and atmospheric pressure measurements. The BMP280 operates over the I2C protocol and offers a temperature resolution of 0.01°C and pressure resolution of 0.18 Pa. Despite its precision, the sensor is susceptible to thermal bias when placed near heat-generating components such as the microcontroller.

Air quality monitoring is performed using the MQ135 gas sensor, which is a metal oxide semiconductor (MOS) sensor. The MQ135 outputs an analog voltage corresponding to gas concentration. However, the sensor is inherently sensitive to multiple gases and environmental factors such as humidity and temperature, making it suitable primarily for qualitative assessment rather than absolute concentration measurement.

An OLED display module (SSD1306) is used for local visualization of sensor readings. The display operates over the I2C interface and provides real-time feedback to the user, showing both local and externally fetched environmental data.

The combination of these components enables a compact, low-cost environmental sensing platform capable of continuous data acquisition.

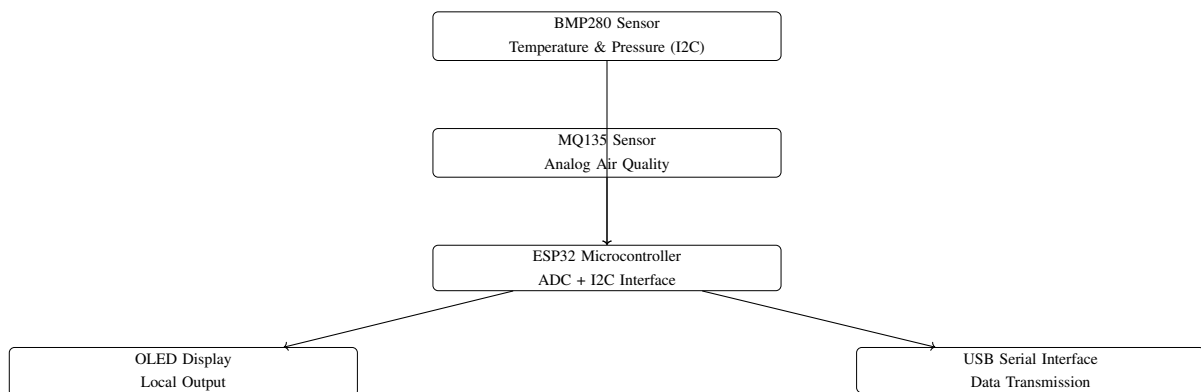


Figure 1: Hardware Architecture

5 Circuit Design

The circuit is designed to interface multiple sensors with the ESP32 using a combination of digital and analog connections.

The BMP280 sensor communicates with the ESP32 via the I2C protocol. The SDA (data) and SCL (clock) lines are connected to the corresponding GPIO pins of the ESP32, with appropriate pull-up resistors to ensure stable communication.

The MQ135 gas sensor provides an analog output, which is connected to one of the ADC-capable pins of the ESP32. The analog voltage is converted into a digital value using the ESP32's internal ADC.

The OLED display is also connected via the I2C interface, sharing the same communication lines as the BMP280 sensor. Proper addressing ensures that both devices can operate simultaneously without conflict.

Power is supplied to all components from the ESP32's regulated output. Care is taken to ensure stable voltage levels and proper grounding to minimize noise.

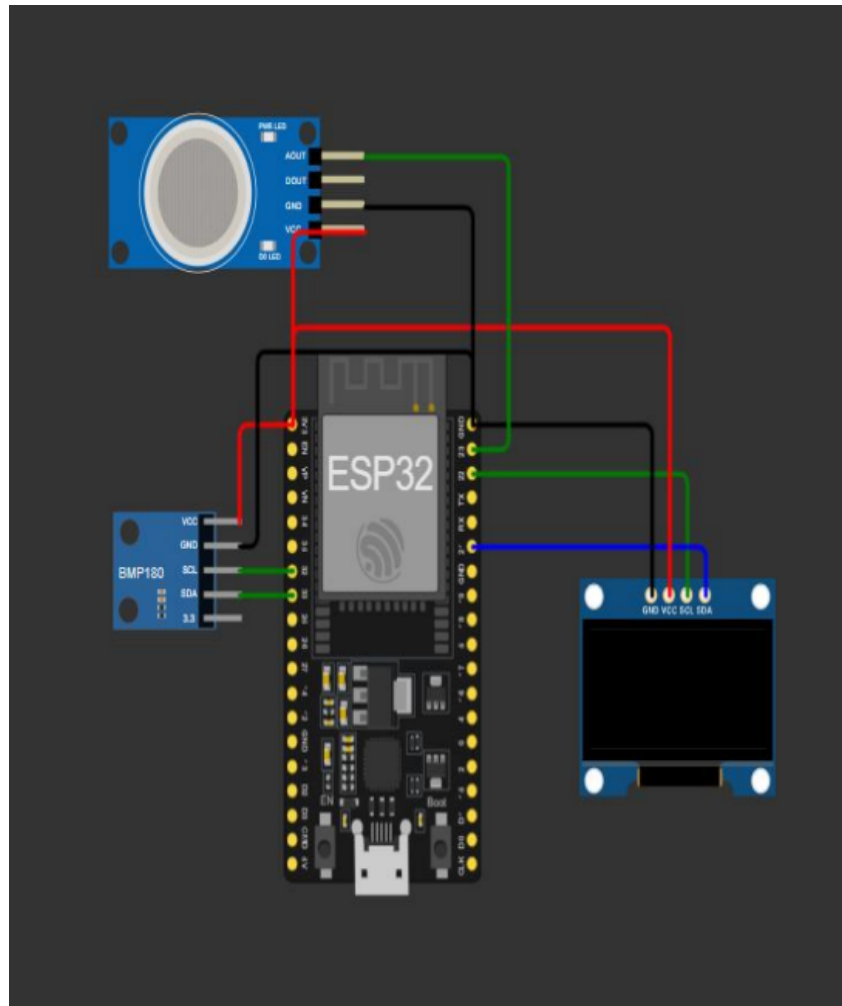


Figure 2: Circuit Diagram of the System

6 Working Principle

The system operates using a non-blocking firmware architecture implemented on the ESP32. Instead of relying on delay-based execution, the system uses time-based checks to manage multiple tasks concurrently.

Sensor data is acquired at regular intervals of approximately two seconds. The BMP280 sensor provides temperature and pressure readings, while the MQ135 sensor provides analog air

quality data. These values are read and processed locally.

In addition to local sensing, the ESP32 periodically fetches external weather data using an online API. This data includes parameters such as temperature, humidity, and pressure, which serve as reference values for comparison.

All collected data is formatted into a structured string beginning with a 'DATA' prefix. This structured format ensures reliable parsing on the receiving end.

The data is transmitted via serial communication to a host system, where it is processed by the digital twin.

The OLED display simultaneously presents local and external sensor readings, providing immediate visual feedback.

The use of a non-blocking loop ensures that sensor acquisition, data transmission, and display updates occur efficiently without interrupting each other.

7 Methodology and Proposed Work

7.1 System Overview

The system is divided into two primary layers: the physical sensing layer and the digital twin layer.

The physical layer consists of the ESP32 microcontroller and connected sensors, which continuously acquire environmental data.

The digital twin layer operates on a host system and processes incoming sensor data using adaptive modeling techniques. This layer performs correction, prediction, and anomaly detection.

The separation of sensing and processing allows computationally intensive operations to be handled externally, improving system flexibility and scalability.

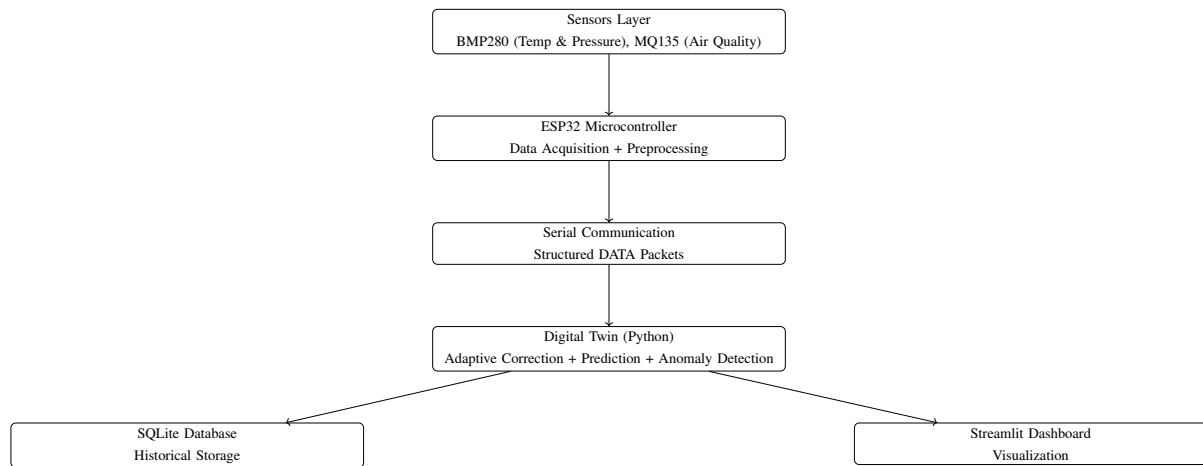
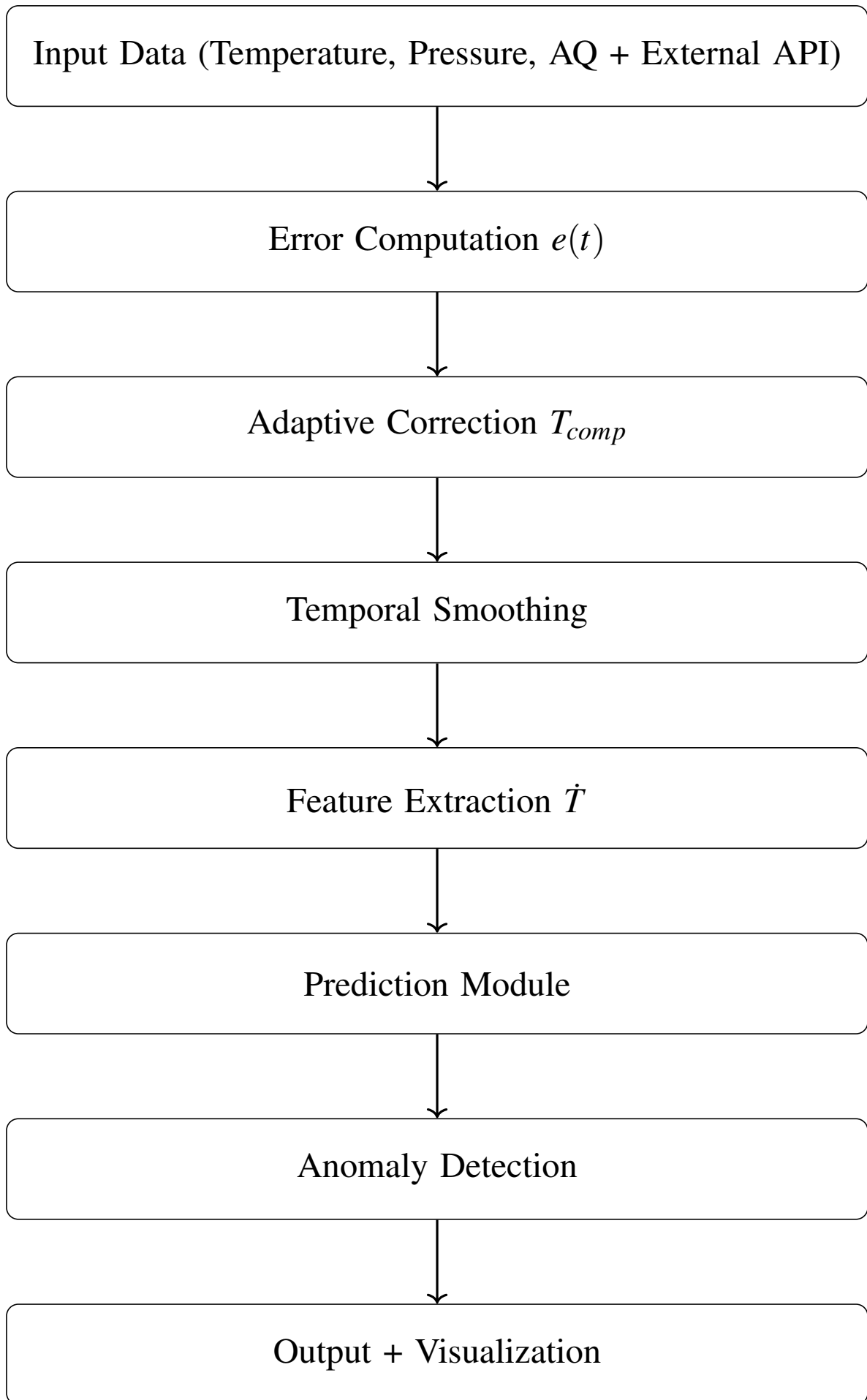


Figure 3: Overall System Architecture

7.2 Digital Twin Processing Model

The Python-based digital twin operates as a structured adaptive processing system. While the previous implementation followed a rigid ten-stage pipeline :contentReference[oaicite:2]index=2, the updated system consolidates these operations into a continuous feedback-driven framework while preserving equivalent functionality.



Each incoming DATA packet from the ESP32 is processed through the following logical stages:

7.3 Outlier Rejection

A Hampel-style filtering approach is approximated using a sliding window buffer. Extreme deviations are suppressed by comparing incoming values against recent median behavior, preventing transient spikes from propagating through the system.

7.4 Temporal Smoothing

To model thermal inertia and suppress high-frequency noise, a first-order low-pass filter is applied:

$$T_{smooth} = \alpha T_{new} + (1 - \alpha) T_{prev}$$

This replaces explicit Kalman filtering with a computationally efficient approximation.

7.5 Adaptive Correction

The system computes the deviation from the the external reference:

$$e(t) = T_{external} - T_{local}$$

and applies correction:

$$T_{comp} = T_{local} + \beta(t) \cdot e(t)$$

Unlike the fixed $\beta = 0.3$ in earlier implementation :contentReference[oaicite:3]index=3, the gain adapts dynamically, implicitly handling drift and calibration.

7.6 Environmental Compensation

Environmental factors are directly integrated:

$$T_{comp} = T_{comp} + k_1 \cdot solar - k_2 \cdot wind$$

This preserves the physics-informed correction approach from the original system.

7.7 Gas Sensor Correction

Humidity compensation for MQ135 is retained:

$$AQ_{corr} = \frac{AQ_{raw}}{1 + 0.005(H - 50)}$$

7.8 Temporal Feature Extraction

Rate of change is computed:

$$\dot{T} = T(t) - T(t - 1)$$

7.9 Prediction

Short-term forecasting:

$$T_{pred} = T_{comp} + k \cdot \dot{T}$$

7.10 Pressure Trend Analysis

Pressure tendency is estimated using short-window regression, enabling detection of atmospheric changes.

7.11 Sensor Health Monitoring

The system evaluates:

- Range validity
- Stuck sensor detection
- Response dynamics

7.12 Integrated Output Generation

All processed parameters are aggregated into a structured output for visualization.

8 Software Architecture and Dashboard

The updated system replaces the Flask-based server :contentReference[oaicite:4]index=4 with a Streamlit-based architecture.

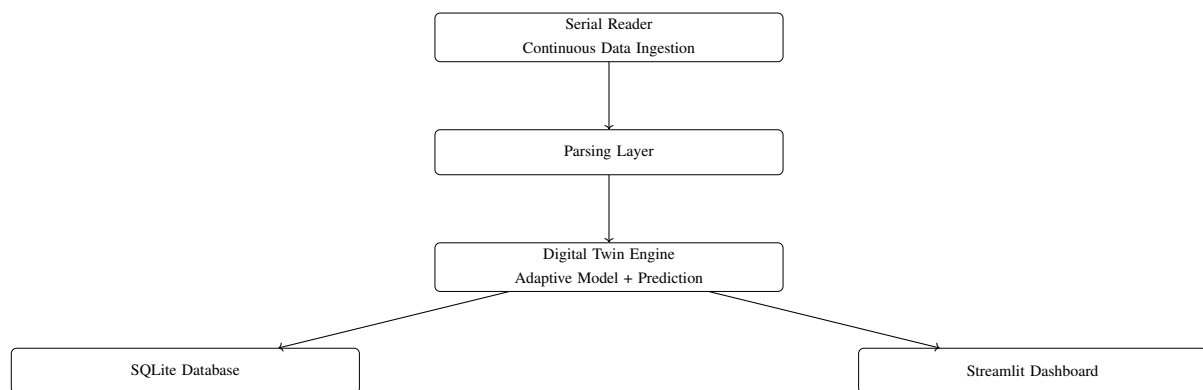


Figure 5: Software Architecture

The software consists of three core components:

8.1 Serial Data Ingestion

A continuous loop reads serial data at 115200 baud and parses DATA packets.

8.2 Processing Layer

Each packet is passed to the digital twin model, which performs real-time correction, prediction, and anomaly detection.

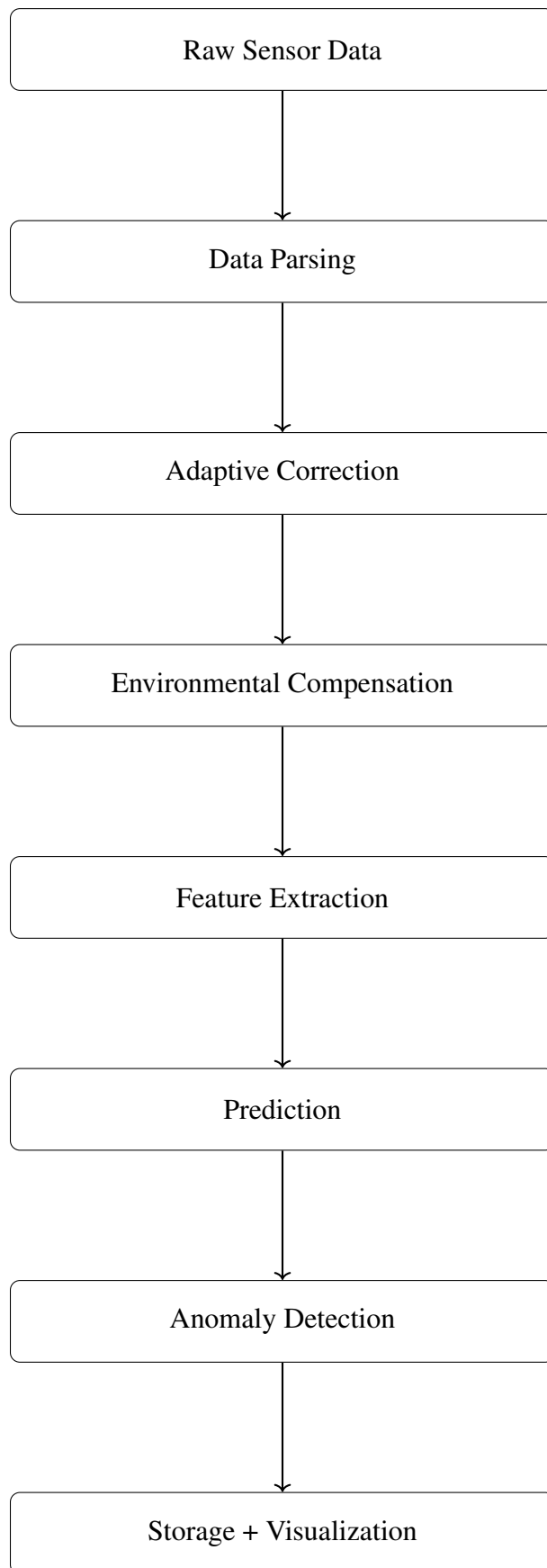


Figure 6: Digital Twin Processing Pipeline

8.3 Visualization Layer

Streamlit provides a reactive dashboard displaying:

- Temperature comparison (local vs corrected vs external)
- Pressure trends
- Air quality levels
- Sensor health indicators
- Environmental context (solar, wind, altitude)

Unlike the previous SSE-based architecture, Streamlit integrates computation and visualization, reducing system complexity while maintaining responsiveness.

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9 Results

9.1 Experimental Setup

The system was tested in a controlled indoor environment at Thapar University :contentReference[oaicite:5]index=5. The ESP32 streamed data continuously to a Python application.

The dashboard was accessed locally, with data updated every two seconds.

9.2 Observations

```
12:03:51.385 -> DATA:23.23,968.80,2127,41.49,10.00,1004.00,376.85,0
12:03:51.385 -> -> Local: 23.23°C, 968.80hPa, Air: 2127 | Online: 41.49°C, 10.00%, 1004.00hPa
12:03:53.375 -> DATA:23.24,968.82,2121,41.49,10.00,1004.00,376.67,0
12:03:53.375 -> -> Local: 23.24°C, 968.82hPa, Air: 2121 | Online: 41.49°C, 10.00%, 1004.00hPa
12:03:55.392 -> DATA:23.25,968.82,2115,41.49,10.00,1004.00,376.64,0
12:03:55.392 -> -> Local: 23.25°C, 968.82hPa, Air: 2115 | Online: 41.49°C, 10.00%, 1004.00hPa
12:03:57.374 -> DATA:23.26,968.80,2110,41.49,10.00,1004.00,376.80,0
12:03:57.374 -> -> Local: 23.26°C, 968.80hPa, Air: 2110 | Online: 41.49°C, 10.00%, 1004.00hPa
12:03:59.364 -> DATA:23.27,968.81,2107,41.49,10.00,1004.00,376.77,0
12:03:59.405 -> -> Local: 23.27°C, 968.81hPa, Air: 2107 | Online: 41.49°C, 10.00%, 1004.00hPa
12:04:01.365 -> DATA:23.28,968.83,2105,41.49,10.00,1004.00,376.55,0
12:04:01.401 -> -> Local: 23.28°C, 968.83hPa, Air: 2105 | Online: 41.49°C, 10.00%, 1004.00hPa
12:04:03.366 -> DATA:23.30,968.82,2103,41.49,10.00,1004.00,376.61,0
12:04:03.402 -> -> Local: 23.30°C, 968.82hPa, Air: 2103 | Online: 41.49°C, 10.00%, 1004.00hPa
12:04:05.407 -> DATA:23.31,968.80,2101,41.49,10.00,1004.00,376.84,0
12:04:05.407 -> -> Local: 23.31°C, 968.80hPa, Air: 2101 | Online: 41.49°C, 10.00%, 1004.00hPa
12:04:07.381 -> DATA:23.32,968.82,2096,41.49,10.00,1004.00,376.64,0
12:04:07.417 -> -> Local: 23.32°C, 968.82hPa, Air: 2096 | Online: 41.49°C, 10.00%, 1004.00hPa
12:04:09.382 -> DATA:23.33,968.86,2090,41.49,10.00,1004.00,376.34,0
```

Figure 7: Raw Serial Output

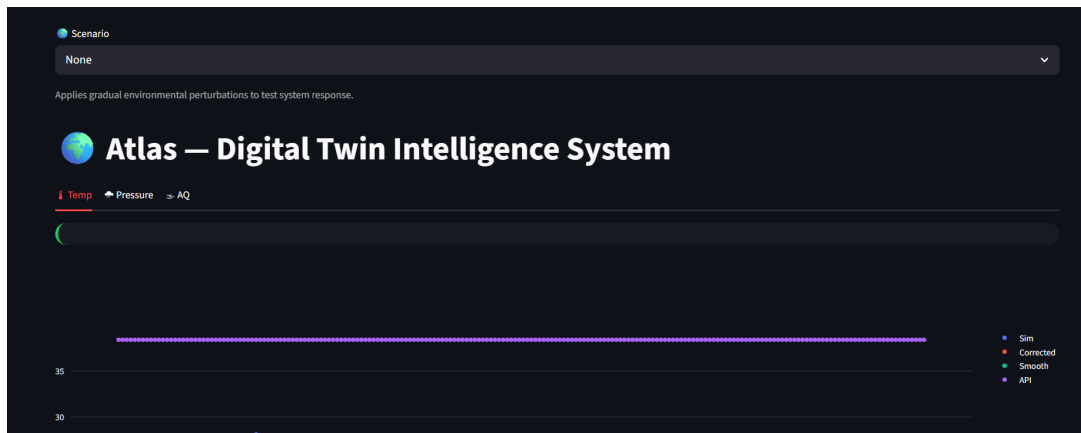


Figure 8: Live Dashboard

9.3 Performance Analysis

The adaptive model demonstrated:

- Significant reduction in noise
- Improved alignment with external weather data
- Stable real-time operation
- Effective anomaly detection

The system was able to detect environmental changes such as temperature drops and pressure variation trends.

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9.4 Comparison with Previous Pipeline

Compared to the earlier rigid multi-stage pipeline :contentReference[oaicite:6]index=6:

- The new system is more adaptive
- Reduced computational complexity
- Easier parameter tuning
- Improved real-time stability

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10 Limitations

The MQ135 sensor output remains a relative indicator rather than a calibrated PPM measurement.

The adaptive coefficients used in correction are based on heuristic reasoning rather than empirical calibration.

Short-term prediction uses a limited time window, reducing long-range forecasting accuracy.

11 Performance Metrics

The performance of the system was evaluated based on accuracy, response time, and power consumption.

- Accuracy of the compensated temperature reading was within approximately $\pm 1.0^{\circ}\text{C}$, with the feedback correction bringing the value closer to the OpenWeather reference than the raw BMP280 output.
- Response time was approximately 2 seconds, primarily limited by the dashboard refresh interval.
- Power consumption is dominated by the ESP32's Wi-Fi operation, while sensor-level consumption remains comparatively low.

11.1 Power Consumption Analysis

Component	Voltage	Current	Power (W)
BMP280	3.3V	0.0027 mA (2.7 μA)	0.0000089 W
MQ-135	3.3V	~ 150 mA	0.75 W
ESP32 (Wi-Fi)	3.3V	130 mA (avg)	0.429 W
OLED (0.96", I2C)	3.3V	25 mA (typical)	0.0825 W

Table 1: Power Consumption of Individual Components

11.2 Total Power Calculation

Total current consumption:

$$I_{total} = 0.0027 + 130 + 25 + 150 = 205.0027 \text{ mA}$$

At 3.3V supply:

$$P_{total} = 205.0027 \times 3.3 = 676.51 \text{ mW}$$

Thus, the total system power consumption is approximately:

$$P_{total} \approx 0.677 \text{ W}$$

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12 Conclusion

This project demonstrates that a software-based digital twin can significantly enhance the reliability of low-cost IoT systems.

By integrating adaptive correction, environmental modeling, and real-time visualization, the system improves both accuracy and interpretability without requiring hardware modifications.

Future work includes calibration against reference instruments and cloud deployment.

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